DeepSPINE: Automated Lumbar Vertebral Segmentation, Disc-Level Designation, and Spinal Stenosis Grading Using Deep Learning

Authors:	Lu
Publication / Venue:	Machine Learning for Healthcare
Year:	2018
DOI:	N/A
Primary Source URL:	https://arxiv.org/abs/1807.10215
Inventory Category:	DL stenosis / multi-pathology
Comparability / Priority:	Very High | Priority: Critical

Study Details & Methodology from Literature Inventory

Study Type / MRI View:	Segmentation + localization + grading | Sagittal and axial T2-weighted MRI
Dataset & Cohort:	Large clinical lumbar MRI cohort (Sample: 4,075 patients; >22,000 disc-level annotations)
Disease / Target:	Central and foraminal lumbar stenosis
Model / AI Method:	U-Net + multi-input / multi-task CNN
Evaluation Metrics:	Segmentation, localization and classification metrics
Key Finding / Relevance:	None
Thesis Context (Ch 2):	A major anatomy-aware, multi-view predecessor; directly challenges a raw image-level classification framing.

Abstract / Summary

No abstract could be retrieved automatically from open-access indexes. Full text remains behind publisher paywall; please see primary source URL above.